

Assignment 3: Data Exploration

Sadie DiCarlo

OVERVIEW

This exercise accompanies the lessons in Environmental Data Exploration M3: Data Exploration.

Directions

1. Rename this file `<FirstLast>_A03_DataExploration.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Assign a useful **name to each code chunk** and include ample **comments** with your code.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.
7. After Knitting, submit the completed exercise (PDF file) to the dropbox in Canvas.

TIP: If your code extends past the page when knit, tidy your code by manually inserting line breaks.

TIP: If your code fails to knit, check that no `install.packages()` or `View()` commands exist in your code.

Set up your R session

1. Load necessary packages (tidyverse, lubridate, here), check your current working directory and upload two datasets: the ECOTOX neonicotinoid dataset (ECOTOX_Neonicotinoids_Insects_raw.csv) and the Niwot Ridge NEON dataset for litter and woody debris (NEON_NIWO_Litter_massdata_2016-26_raw.csv). Name these datasets “Neonics” and “Litter”, respectively.

```
library(tidyverse);  
library(lubridate);  
library(here)  
  
getwd()
```

```
## [1] "/home/guest/EDE_R_Lab_F26"
```

```
Neonics <- read.delim( #delim because its a txt file  
  file = here("Lessons", "ECOTOX_Neonicotinoids_Insects_raw.txt"),  
  sep = "|", #separate by | instead of commas  
)
```

```
Litter <- read.csv(
  file = here('Lessons', 'NEON_NIWO_Litter_massdata_2016-26_raw.csv')
)
```

```
dim(Neonics) #dimesions of Neonics as a check that separation worked
```

```
## [1] 8748 76
```

Learn about your system

2. The neonicotinoid dataset was collected from the Environmental Protection Agency's ECOTOX Knowledgebase, a database for ecotoxicology research. Neonicotinoids are a class of insecticides used widely in agriculture. The dataset that has been pulled includes all studies published on insects. Why might we be interested in the ecotoxicology of neonicotinoids on insects? Feel free to do a brief internet search if you feel you need more background information.

Answer: We might want to know what how effective the insecticide is and the broader effects of the insecticides are. Would they impact just the target insects or could they endanger other insects that are crucial to the ecosystem. Also we could be interested in neonicotinoids effect living things broadly and we could be interested in their effect on humans.

3. The Niwot Ridge litter and woody debris dataset was collected from the National Ecological Observatory Network, which collectively includes 81 aquatic and terrestrial sites across 20 ecoclimatic domains. 32 of these sites sample forest litter and woody debris, and we will focus on the Niwot Ridge long-term ecological research (LTER) station in Colorado. Why might we be interested in studying litter and woody debris that falls to the ground in forests? Feel free to do a brief internet search if you feel you need more background information.

Answer: We could be interested in which plants are shedding and what they are shedding. Quantifying litter might give a rough estimate of how many of certain species are in the forest. We could also use this to learn about how litter becomes incorporated into soil.

4. How is litter and woody debris sampled as part of the NEON network? Read the NEON_Litterfall_UserGuide.pdf document to learn more. List three pieces of salient information about the sampling methods here:

Answer: 1.Sampling is conducted over the lifetime of the Observatory. 2. Sampling plots are randomly selected within the primary and secondary watersheds. 3. One to four elevated or ground traps for every 400 m² plot

Obtain basic summaries of your data (Neonics)

5. What are the dimensions of the dataset?

```
dim(Neonics) # rows, then columns of Neonics dataset
```

```
## [1] 8748 76
```

6. Using the `summary` function on the “Effect” column, determine the most common effects that are studied. Why might these effects specifically be of interest? [Tip: The `sort()` command is useful for listing the values in order of magnitude...]

```
Neonics$Effect <- as.factor(Neonics$Effect) #change to factor
sort(summary(Neonics$Effect), decreasing = TRUE) #return most common categories of effects
```

##	Mortality	Population	Genetics	Behavior
##	2687	1377	905	826
##	Feeding behavior	Reproduction	Development	Enzyme(s)
##	712	557	499	405
##	Growth	Biochemistry	Morphology	Avoidance
##	153	151	87	86
##	Physiology	Cell(s)	Intoxication	Accumulation
##	79	57	52	47
##	Immunological	Histology	Hormone(s)	Injury
##	38	20	8	2

Answer: The most common effects studied include Mortality, Population, and Genetics which would be very important if you want to understand how this chemical is affecting the populations and their ability to sustain themselves.

7. Using the `summary` function, determine the six most commonly studied species in the dataset (common name). What do these species have in common, and why might they be of interest over other insects? Feel free to do a brief internet search for more information if needed. [TIP: Explore the help on the `summary()` function, in particular the `maxsum` argument...]

```
Neonics$Species.Scientific.Name. <- as.factor(Neonics$Species.Scientific.Name.) #change to factor
summary(Neonics$Species.Scientific.Name., maxsum = 7) #list most common scientific names studied
```

##	Apis mellifera	Apis mellifera ssp. carnica
##	2310	622
##	Bombus impatiens	Apis mellifera ligustica
##	357	348
##	Coccinella septempunctata	Bombus terrestris
##	270	216
##	(Other)	
##	4625	

Answer: The most commonly studied species are bees and beetles. It would be very important to study pollinators in order to understand the effect of the insecticide on the rest of the ecosystem and make sure that only specific pest insects are being affected.

8. Concentrations are always a numeric value. What is the class of `Conc.1.Mean..Author..` column in the dataset, and why is it not numeric? [Tip: Viewing the dataframe may be helpful...]

```
class(Neonics$X.Conc.1.Mean..Author..) #see class of column
```

```
## [1] "character"
```

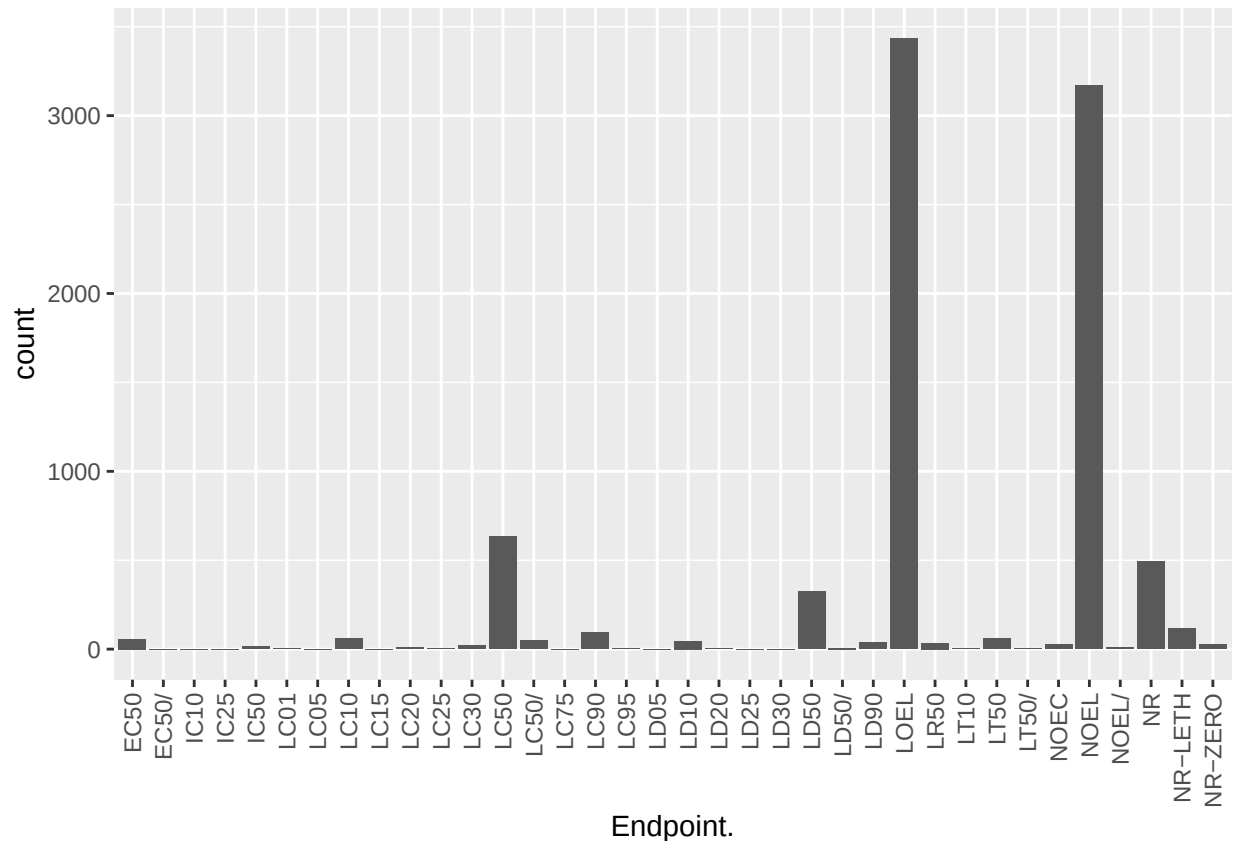
Answer: It's class is characters because you're not using these numbers for math. Instead they are identifiers.

Explore your data graphically (Neonics)

9. Create a bar graph of Endpoint counts. What are the two most common end points, and how are they defined? Consult the ECOTOX_CodeAppendix for more information.

[**TIP:** Add `theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))` to the end of your plot command to rotate and align the X-axis labels...]

```
ggplot(Neonics, aes(x = Endpoint.)) +  
  geom_bar() + theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))
```



```
#graph endpoints of Neonics dataset
```

Answer: The two most common endpoints are LOEL and NOEL which stand for Lowest Observed effect level and No Observed effect level.

Explore your data (Litter)

10. Determine the class of collectDate. Is it a date? If not, change to a date and confirm the new class of the variable. Using the `unique` function, determine which dates litter was sampled in August 2018.

```
class(Litter$collectDate) #determine class
```

```
## [1] "character"
```

```
Litter$collectDate <- as.Date(Litter$collectDate, format = "%Y-%m-%d") #change format
class(Litter$collectDate) #return class
```

```
## [1] "Date"
```

```
length(unique(Litter$collectDate)) #return number of unique dates in column
```

```
## [1] 56
```

11. Using the unique function, list the different plotIDs sampled at Niwot Ridge. How is the information obtained from unique different from that obtained from summary?

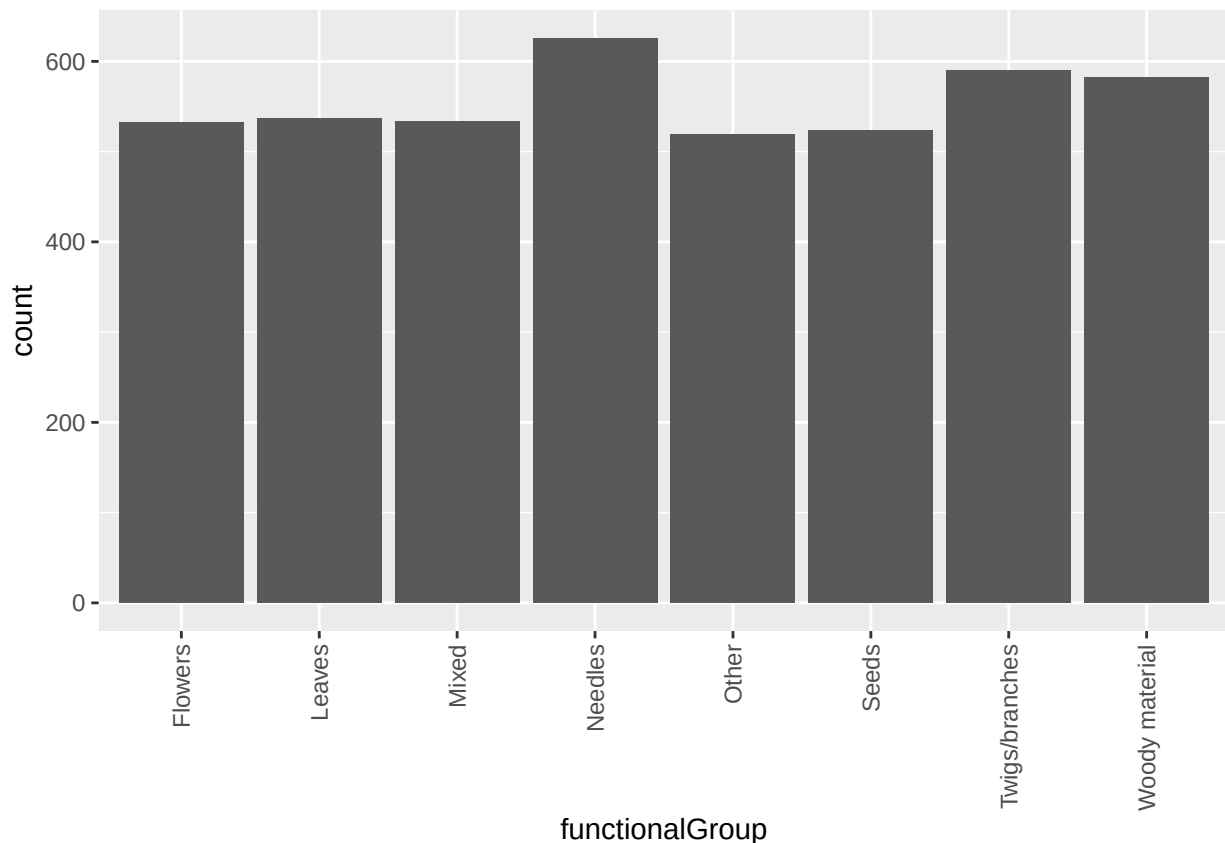
```
length(unique(Litter$plotID)) #return number of unique plotIDs
```

```
## [1] 12
```

Answer: Unique tells you how many different plotID values there are. Summary just tells you how many pieces of data are under plotID column.

12. Create a bar graph of functionalGroup counts. This shows you what type of litter is collected at the Niwot Ridge sites. Notice that litter types are fairly equally distributed across the Niwot Ridge sites.

```
ggplot(Litter, aes(x = functionalGroup)) +
  geom_bar() + theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))
```



```
#plot groups of litter biomass
```

What type(s) of litter tend to have the highest biomass at these sites?

Answer: Needles

13. Write a short data-exploration summary for Neonics.

- **One key characteristic:** In most insects studied the levels of chemical contamination is low to non-existent.
- **One data-quality issue:** Many variables are in the wrong class to be used and need to be changed in order to be included in r plots.
- **One interesting pattern:** Pollinators were the most commonly studied insect in this study.
- **One question for further analysis:** Which insects have high levels of chemicals in them.

14. Write a short data-exploration summary for Litter.

- **One key characteristic:** Needles are the group with the highest biomass of all the litter in the plots.
- **One data-quality issue:** There are multiples of the same data point in some of the categories and you have to determine if you need to group or not.
- **One interesting pattern:** There are relatively similar amounts of litter biomass between the litter groups.
- **One question for further analysis:** temporal analysis of the litter looking at how litter biomass changes across seasons and years.